Paleomagnetism of the Flin Flon – Snow Lake Greenstone Belt, Manitoba and Saskatchewan[1]

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The Flin Flon – Snow Lake greenstone belt is of Archean or Aphebian age and has been metamorphosed to greenschist or low amphibolite facies during the Hudsonian orogeny. It contains four rock units — the Amisk Group, the Missi Group, metadiorite and metagabbro bodies, and the Boundary Intrusions. The rocks contain two main magnetizations, an early **B** magnetization and a later **A** magnetization. Both magnetizations contain reversals. The **A** magnetization (43 sites, direction 168°, +73°, $k = 17$, $\alpha_{95} = 5°$, pole 24°N, 095°W) is found in all rock units. It can be divided into three parts (1, 2 and 3) which have the same general direction, but which are significantly different from one another. It is suggested that **A1** is a viscous partial thermoremanence (vptrm), and **A2** possibly a chemical remanence (crm). Both are considered to have been acquired during uplift (1600–1700 m.y.) following the Hudsonian orogeny. **A3** is possibly a crm acquired during late fault movements and appears to be carried by hematite developed during retrograde metamorphism. **A3** is considered to be *about* 1600 m.y. old. The **B** magnetization (7 sites, 129°, +04°, $k = 12$, $\alpha_{95} = 18°$, pole 20°S, 046°W) occurs in all rock units except the Missi. It is particularly well developed in the Boundary Intrusions where it is considered to be a primary thermoremanent magnetization (trm). In other rock units **B** is considered to be a vptrm acquired during heating at the time of emplacement of the Boundary Intrusions. The **B** magnetization is considered to be *about* 1800 m.y. old. Some enigmatic magnetizations with very high blocking temperatures above 700°C were observed. Formulas for calculating the coercivities in rocks with more than one magnetization are given.

La ceinture de greenstones Flin Flon – Snow Lake est d'âge Archéen ou Aphébien, et a été métamorphisée au faciès schistes verts ou à amphibolite peu élevé durant l'orogène Hudsonnienne. Elle contient trois unités lithologiques—le Groupe Amisk, le Groupe Missi, des intrusions de métadiorite et de métagabbro et les "Boundaries Intrusions" (Intrusions Bordières). Les roches contiennent deux magnétisations principales, une première magnétisation **B**, et une deuxième magnétisation **A** subséquente. Les deux magnétisations contiennent des inversions. La magnétisation **A** (43 sites, directions 168°, +73°, $k = 17$, $\alpha_{95} = 5°$, pôle 24°N, 095°W) se retrouve dans toutes les unités lithologiques. Elle peut être divisée en trois parties (1.2 et 3) qui ont la même direction générale mais sont quand même assez différentes l'une des autres. Il est suggéré que **A1** est une thermorémanence partielle visqueuse, (vptrm), et **A2** une remanence chimique possiblement (crm). Les deux sont considérées comme acquises durant soulèvement (1600–1700 m.a.) après l'orogène Hudsonnienne. **A3** est peut-être une rcm acquise durant les mouvements tartifs de failles et semble être portée par de l'hématite développée durant un métamorphisme retrograde. On condisère que **A3** a un âge d'*environ* 1600 m.a. La magnétisation **B** (7 sîtes, 129°, +04°, $k = 12$, $\alpha_{95} = 18°$, pôle 20°S, 046°W) se retrouve dans toutes les unités sauf dans le Missi. Elle est particulièrement bien développée dans les "Boundary Intrusions" où elle est considérée comme une magnétisation thermoremanente primaire (trm). Dans les autres unités, **B** est considérée comme une trmpv acquise durant le réchauffement où l'époque de la mise en place des "Boundary Intrusions". La magnétisation a *environ* 1800 m.a. Certaines magnétisations énigmatiques avec de très hautes températures de démagnétisation au-dessus de 700°C ont été observées. On donne formules pour calculer les coercivités dans les roches ayant plus d'une magnétisation.

[Traduit par le journal]

Introduction

The Flin Flon – Snow Lake greenstone belt extends across central Saskatchewan and Manitoba (Figs. 1 and 2). It is a major mining region yielding base metals (Cu, Pb, Zn) and byproducts (Au, Ag, Cd, Se) from massive sulfide ores. The rocks are metamorphosed to greenschist or amphibolite facies. Numerous geological studies have been made but this is the first dealing with paleomagnetism. Paleomagneticians have usually tended to shun metamorphic terrains in favor of less altered rocks whose magnetism could be

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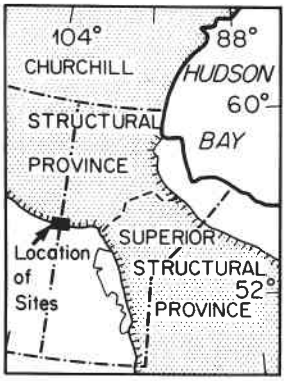


FIG. 1. General sampling area (see Fig. 2). The Canadian Shield is stippled.

more readily interpreted. Recently I completed a paleomagnetic survey of metamorphic rocks in the northern part of the Churchill Structural Province of the Canadian Shield (Park 1973), and a survey in the southern part of the Province seemed a natural next step to take. The study was generally focused on the easily accessible well-mapped Flin Flon area where unweathered rocks could be sampled at roadside exposures, and where there are numerous radiometric age determinations. These have generally given late Aphebian ages, but a few late Archean dates exist. The question of the age of the greenstone belt is therefore controversial. Are the Archean dates correct and the younger Aphebian values updated, or do the Aphebian ages measure the true age of the belt, the apparent Archean ages being in error? Thus one question that I have had before me during the course of this work is, are pre-Aphebian magnetizations present in these rocks? This has led to an attempt to relate the magnetic record to the deformational phases that have been recognized. To the author's knowledge this has never been attempted in any other paleomagnetic study and therefore the conclusions are necessarily very tentative.

Geology and Geochronology

The Flin Flon – Snow Lake greenstone belt mainly comprises a sequence of metavolcanics (Amisk Group) and metasediments (Missi Group) intruded by acid to ultrabasic rocks (the Boundary Intrusions) (Figs. 2 and 3). Immediately north

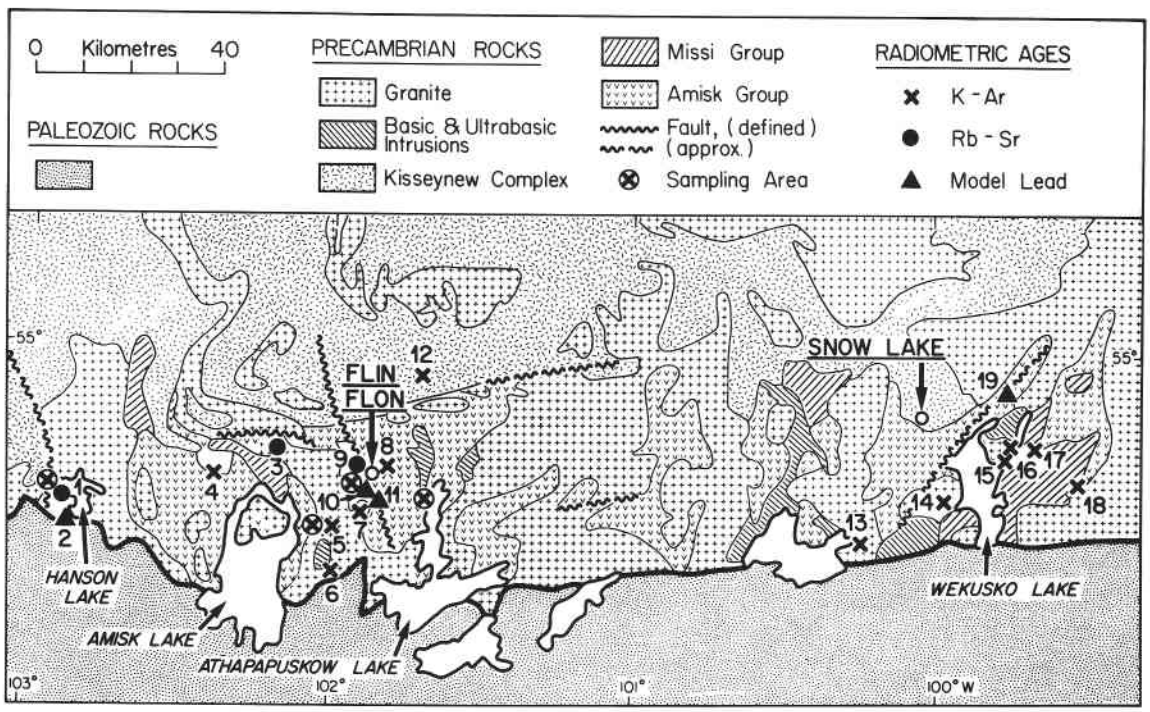


FIG. 2. Flin Flon – Snow Lake greenstone belt showing generalized geology (after Geol. Surv. Canada Map 1164 A), the paleomagnetic sampling areas, and age sampling localities. For age data see Table 1.

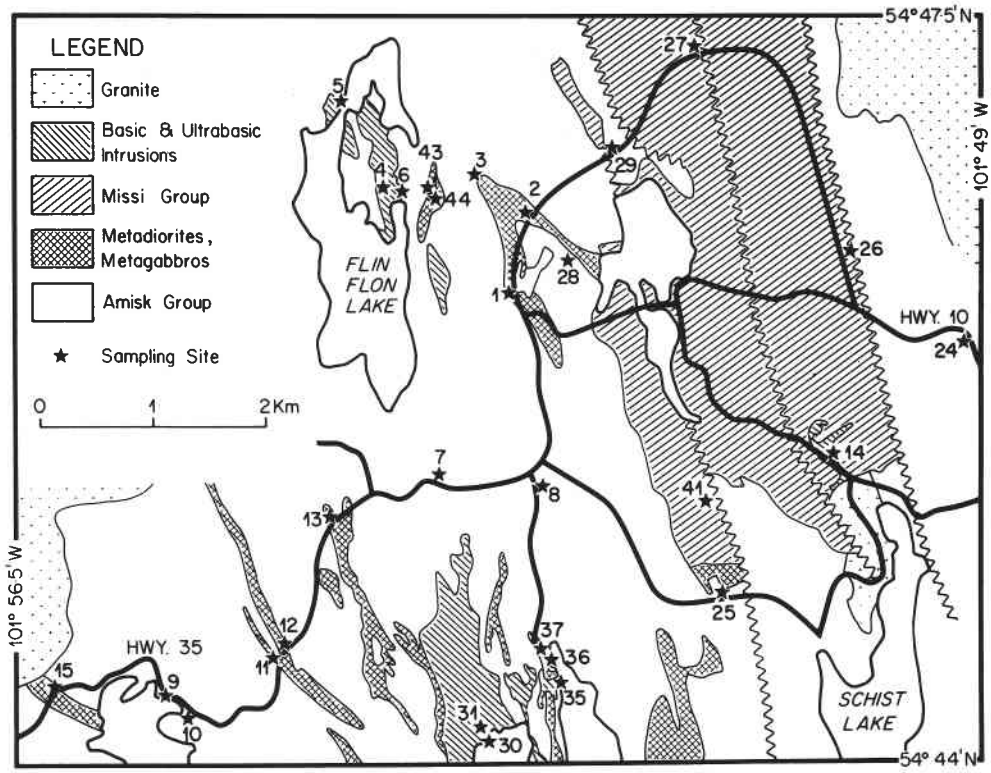


FIG. 3. Generalized geology at Flin Flon after Stockwell (1960) and Sask. Dep. of Mineral Res. Map 62C. The Basic and Ultrabasic bodies are the Boundary Intrusions.

and west lies the Kisseynew Complex, which may be the highly metamorphosed equivalent of the Amisk and Missi. The Wasekwan and Sickle Groups, found in the Lynn Lake area, 208 km north of Flin Flon, may also be related to the Amisk and Missi Groups. Sangster (1972) has suggested that all groups comprise the eugeosynclinal equivalents of the miogeosynclinal Hurwitz Group which occurs still further to the north.

Many workers have contributed to the geology of the Flin Flon and Amisk Lake areas (see Byers and Dahlstrom 1954; Byers *et al.* 1965; Stauffer and Mukherjee 1971; and references therein). Their work has been reviewed by Bailes (1971). Much of the mapping near Flin Flon was done by Stockwell (1960). The Amisk Group, the oldest unit at Flin Flon, consists of fine-grained basalts and andesites, with subordinate dacites, rhyolites, and quartz porphyry. There are numerous metadiorite and metagabbro bodies that are probably the intrusive equivalents of the Amisk lavas. The Amisk tends to become more basic lower in the section near Amisk Lake (Fig. 2). Amisk rocks have been sampled at 16 sites, and all but one (site 25) are in the basic lavas of map unit 1 of Stockwell (1960) (Fig. 3). The metadiorites and metagabbros have been sampled at 11 sites, four of them near Amisk Lake.

The Missi Group, although resting unconformably on the Amisk, was probably deposited shortly after the end of Amisk volcanism (Byers *et al.* 1965). It comprises a basal conglomerate and a sequence of cross-bedded subarkoses and feldspathic greywackes derived mainly from Amisk material (Stauffer and Mukherjee 1971). Missi rocks within about 100 m of faults have a red iron-staining due to hematite. Seven sites were collected from the Missi Group. Three are from greyish-green, fine-grained rocks near Flin Flon (Fig. 3), and four are from red, fine to medium grained rocks close to fault zones near the airport east of the town (Fig. 2).

The Amisk and Missi Groups have been intruded by granitic and basic bodies which range from Amisk to post-Missi in age. The oldest intrusive rocks appeat to be minor sills and dikes,

and the metadiorites and metagabbros mentioned above. The next intrusive phase consists of plutons and smaller dikes of granite and granodiorite which cut both Amisk and Missi rocks. These in turn are cut by a series of acidic to ultrabasic intrusions. The basic and ultrabasic bodies are called the Boundary Intrusions (Fig. 3). Nine sites were sampled.

About 40 km west of Flin Flon in the Hanson Lake area (Fig. 2) Amisk-type metavolcanic and metasedimentary rocks overlie with apparent conformity Kisseynew-type rocks (Coleman *et al.* 1970). Smith (*ibid.*) believes that Amisk-type rocks were extruded onto Kisseynew-type rocks which had been previously deformed, metamorphosed, and eroded. The base of the Amisk-type section comprises mafic volcanics overlain by intermediate and felsic volcanics and soda-rhyolite porphyry. The latter may be wholly or partly intrusive. The top of the section, possibly separated from the porphyries by a minor unconformity (Smith, *ibid.*), comprises a sedimentary sequence, the upper part of which is mainly metagreywacke. Three sites were collected from the mafic volcanic rocks near Hanson Lake (Fig. 2).

The Amisk volcanics appear to be submarine. After possible minor folding, faulting, uplift, tilting and erosion, the Missi Group was deposited in a fluviatile environment. According to Stauffer and Mukherjee (1971) three phases of deformation followed. The first (P1) probably involved the Amisk and Missi Groups only. During the second phase (P2) various syntectonic granitic and granodioritic bodies were also deformed. Prior to phase three (P3) more granitic and granodioritic bodies were intruded. P3 includes faulting and the development of large open folds. The Boundary intrusions are considered to have been intruded either during or after the later stages of P3, since they cut, and are cut by, late stage faults (Byers *et al.* 1965). The economic sulfide deposits appear to be related to this fault system (*ibid.*) and possibly originated with the Boundary Intrusions (Stockwell 1960). There is no evidence that the Boundary Intrusions have been subsequently deformed.

Metamorphism occurred during the latter two phases. During P2, lower greenschist to low amphibolite facies metamorphism developed at Flin Flon, the grade increasing to the northwest. The Amisk and Missi collecting sites are in either the biotite or chlorite zones of the greenschist facies. During P3, retrograde metamorphism developed near faults at Flin Flon. The Boundary Intrusions, though not affected by the regional metamorphism, are all altered, but these effects are attributed to autometamorphism. At Hanson Lake regional metamorphism and extensive deformation have affected all rocks except the last granitic intrusives. Retrograde metamorphism is common.

Numerous radiometric ages have been obtained (Fig. 2). The results have been reviewed by Mukherjee *et al.* (1971) and Sangster (1972) and are summarized in Table 1. K–Ar and Rb–Sr ages have been obtained from Amisk and Missi rocks, and from granitic intrusions; and there is no relationship between age and rock type. Lead ages are from sulfide ores. Most studies yield mean ages within the range 1750 to 1900 m.y. The spread of K–Ar ages is 1620 to 1930 m.y. at Flin Flon, and 1610 to 1960 m.y. at Wekusko Lake. The younger K–Ar ages presumably reflect the last stages of tectonic activity, especially cooling following uplift. Thus final uplift may have occurred about 1600 m.y. ago. Model Pb and most Rb–Sr ages range from 1770 to 1900 m.y. This may indicate the approximate time of formation and metamorphism (if the latter occurred soon after formation), or it may indicate only the time of metamorphism, the time of formation being much earlier. The former view is held by Sangster (1972) who argues that the uniformity in composition of leads in the massive sulfide deposits indicates single-stage lead ages and therefore that the ores are coeval with their host rocks, the Amisk Group. However Cumming and Gudjurgis (1973) have recently questioned the validity of single-stage interpretations for model Pb ages, because of the possible homogenization of lead isotope ratios during regional metamorphism. Hence, the ages of metamorphism and formation of these rocks may be very different. This latter view is supported by two Rb–Sr age determinations from Hanson Lake (Coleman 1970). One determination of 2446 ± 16 m.y. is from a granite which intrudes Amisk-like rocks. Coleman favors the idea that these ages indicate the time at which igneous activity occurred rather than the time of a Kenoran metamorphism. The other determination (2521 ± 60 m.y.) is from Amisk-like rocks. Smith (Coleman *et al.* 1970) points out that some of the rocks from which the samples of Amisk-like rocks were taken are anomalously

TABLE 1. Summary of radiometric age determinations (m.y.)

Region	Hudsonian			Pre-Hudsonian
	K–Ar	Rb–Sr*	Model Lead	Rb–Sr*
Hanson Lake	—	1799 ± 2 (1;1wr)	1780 ± 44 (2;1)	2446 ± 16 (1;1wr) 2521 ± 60 (1;1wr)
Flin Flon	1770 ± 56 (4–8;4bi,1mu)	1775 ± 89 (9;1wr) 1805 ± 14 (3;1wr)	1816 ± 32 (10,11;4)	—
Wekusko Lake	1752 ± 39 (13–18;4bi,4mu)	—	1900 ± 44 (19;1)	—
Kisseynew Complex†	1735 (12;1bi)	—	—	—

NOTES: The standard error of the means is given for the Flin Flon model lead age and the K–Ar ages. Reference numbers (Fig. 2) and the number of determinations made on biotite (bi), muscovite (mu) and the whole rock (wr) are given in brackets. Reference numbers are as follows: 1 (Coleman 1970); 2, 10, 11, 19 (Sangster 1972); 3, 9 (Mukherjee *et al.* 1971); 4, 12, 13 (Lowdon 1961: samples 72, 73, 74); 5, 13, 18 (Lowdon *et al.* 1963: samples 112, 118, 119, 120); 6, 7, 8, 14, 15, 16 (Wanless *et al.* 1965: samples 96, 108, 106, 104, 99, 100); 17 (Wanless *et al.* 1967: sample 97); and 10 (Cumming *et al.* 1955, Stacey *et al.* 1969).

*$\lambda^{87}Rb = 1.39 \times 10^{-11} y^{-1}$.

†See Fig. 2.

high in Sr, and could result in an anomalously high age. This could indicate a metamorphism at about 2425 ± 30 m.y. (Coleman 1970). Other rocks, possibly related to the Amisk-Missi Groups, have also yielded older ages. For example, a K–Ar age determination of 2670 m.y. (Leech *et al.* 1963) from metasedimentary rocks of the Sickle Group near Lynn Lake, Manitoba, was used by Stockwell (1963) to argue that the Flin Flon rocks might be Archean.

Several interpretations have been put forward to account for the results reviewed above. (1) The Amisk, Missi, and intrusive rocks are Aphebian and the deformation Hudsonian (Sangster 1972, preferred also by Byers *et al.* 1965 and Mukherjee *et al.* 1971). (2) The Amisk, Missi, and intrusive rocks are Archean, and the deformations are both Hudsonian and Kenoran (Mukherjee *et al.* 1971, preferred by Stockwell 1963). (3) The Amisk is Archean, the Missi is Aphebian, the intrusive rocks are Hudsonian, and the deformation both Hudsonian and Aphebian (Mukherjee *et al.* 1971). (4) The Amisk rocks are both Archean and Aphebian and the deformation Hudsonian (Sangster 1972). Interpretations (1), (2), and (4) have been generally invoked to explain the ages at Hanson Lake. In that region, interpretation (2) is preferred by Byers (1957), Stockwell (1963), and Coleman *et al.* (1970).

Methods

The study of the paleomagnetism of metamorphic rocks is still in its infancy and an adequate methodology has not yet been developed. Major uncertainties are (1) the length and degree of heating during metamorphism and the possibility of changes in magnetic minerals, (2) the uncertainty in referring the direction to the correct horizontal plane, (3) the technical difficulties in resolving the complex magnetization present, and (4) the uncertainties of relating these magnetizations to the age determinations and to the tectonic phases. All problems cannot be solved at once and in this paper only the last two are dealt with.

Samples collected from a single rock exposure are referred to as a site. Usually 5 core samples were drilled from each site and oriented *in situ* by sun and magnetic compasses. Two specimens (diam. 2.50 cm, ht. 2.20 cm) were cut from each core. Specimens were magnetically and thermally demagnetized in apparatus described by Roy *et al.* (1969, 1971, 1973), and measured with a Schonstedt spinner magnetometer. Site cleaning fields and temperatures were determined by a combination of the 'end-point' (As and Zijderveld 1958) and 'least scatter' (Irving *et al.* 1961) methods. Specimen results were combined to form sample means and these are the basic statistical unit used. The most fruitful pro-

TABLE 2. Example of vector subtraction from site 9

Temperature (°C)	(1)		(2)		(3)	
	M	$D°, I°$	M	$D°, I°$	M	$D°, I°$
nrm	4.9	056, +75	—	—	—	—
100	10.2	027, +70	5.4	192, −63	—	—
200	7.0	354, +49	4.2	158, −07	5.0	125, +60
300	6.0	351, +35	4.7	156, +16	6.8	127, +64
400	4.5	004, +08	5.6	169, +47	9.2	137, +78
500	4.8	000, −05	6.7	166, +51	10.3	137, +77
550	4.3	358, −09	6.6	161, +56	10.5	122, +79
600	2.7	323, −05	5.8	118, +59	10.4	073, +71
650	2.2	014, +22	4.2	156, +71	8.9	044, +80

NOTE: The vector differences given under (2) and (3) are the measured vectors (1) subtracted from the nrm and 100°C vectors respectively. M is the intensity in microgauss, and $D°$ and $I°$, the declination and inclination. Column (2) above 300°C, and column (3) show typical A magnetization, SE with positive inclination. Column (1) above 300°C shows typical B magnetization, to the north with low inclination.

cedure for Flin Flon rocks has proved to be alternating field (af) demagnetization followed by thermal demagnetization. Presumably the reason for this is that af demagnetization adequately removes magnetization owing to the present field, but cannot discriminate between other magnetizations of higher stability because their coercive force spectra overlap. This discrimination can however be made thermally because their blocking temperature spectra are distinct.

The combination and analysis of data is not straightforward because two or more magnetizations commonly occur within single specimens. Mean site directions with high precisions and stability were sometimes obtained, which, on further demagnetization, were found to be composed of two or more magnetizations. These magnetizations usually could be resolved by further demagnetization and vector analysis. An example is shown in Table 2, which shows that two distinct magnetizations (referred to as **A** and **B**, see below) are present and can be separated in this way.

Results from certain samples have been rejected. Criteria for rejection are as follows: (1) the intensities were $<0.3 \times 10^{-6}$ microgauss, (2) the difference between specimen directions from the same core was $>50°$, (3) the directions remained close to the present Earth's field during demagnetization, (4) the magnetization direction exhibited generally erratic behavior during demagnetization such that there were no end-points and the magnetization could not be resolved by vector analysis, and (5) the directions at a site were random by the criteria of Vincenz and Bruckshaw (1960) and Watson (1956b).

Remanence Coercivity and Anisotropy

The presence of several magnetizations in individual specimens calls for the identification of the various coercivities present. A useful technique (which is a development of a procedure described by Carmichael (1961)) has been developed using the demagnetization curves of isothermal remanent magnetization (irm) and is described in the Appendix. In Fig. 4(1) the coercivities obtained in this way from Flin Flon rocks are compiled. There is a wide range of values which may be grouped into three ranges: a lcf (low coercive force, 0–200 Oe) range, a mcf (medium coercive force, 200–1000 Oe) range, and a hcf (high coercive force >1000 Oe) range. Commonly two values occur within the mcf range in the same specimen. The ranges below 1000 Oe are probably owing largely to magnetite, and that above 1000 Oe probably to hematite. The mean values for each rock group are given in Table 3.

The effect of anisotropy on the magnetization directions of metamorphic rocks has been studied using the test of Irving and Park (1973). Several specimens are oriented between the poles of an electromagnet so that the field of the magnet is parallel to the dominant magnetization observed in these specimens. An irm is impressed and then demagnetized by alternating fields. Discrepancies between the average of the cleaned irm directions and the applied field are due to anisotropy. The results given in

TABLE 3. Mdf's of irm (Oe)

Entry	Description	lcf	low-mcf	high-mcf	hcf
1.	Amisk Group	130 ± 24(12)	295 ± 43(11)	538 ± 74(7)	3443 ± 3824(8)
2.	Metadiorites, metagabbros	151 ± 5(5)	318 ± 32(7)	565 ± 122(5)	4937 ± 3670(6)
3.	Average of 1 and 2	136 ± 22(17)	304 ± 40(18)	541 ± 95(12)	4083 ± 3694(14)
4.	Amisk? (Hanson Lake)	85 ± 10(3)	300(1)	—	1150(1)
5.	Boundary Intrusions	131 ± 23(9)	330 ± 35(8)	—	3862(1)
6.	Average of 1, 2, 4, 5	129 ± 26(29)	312 ± 40(27)	541 ± 95(12)	3886 ± 3516(16)

NOTES: The mcf range is divided into two parts, since a number of specimens display two components in this range. The number of determinations is in parentheses.

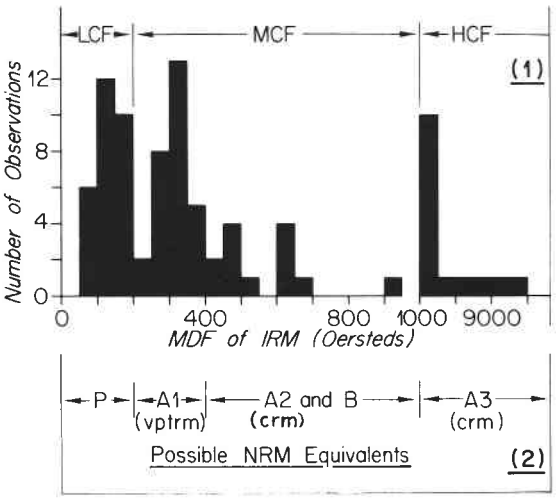


FIG. 4. (1) Mdf of irm spectrum showing the lcf, mcf, and hcf ranges as defined in the text. The spectrum comprises all mdf components in one specimen from each site of the Amisk Group, the metadiorites and metagabbros, the Amisk-like rocks, and the Boundary Intrusions. (2) Possible nrm equivalents of the mdf ranges. *P* is magnetization due to the present field. **A1**, **A2**, and **A3** are phases of the **A** magnetization. Vptrm stands for viscous thermoremanent magnetization and crm for chemical remanent magnetization.

Table 4 show that anisotropy has no significant effect.

TABLE 4. Effect of anisotropy

	Direction (k, α°_{95})
Applied field (18 500 Oe)	184°, +77°
irm, before af cleaning	183°, +77° (104,5°)
irm, after af cleaning	182°, +76° (173,4°)

NOTE: Eleven specimens. k is the precision and α°_{95} the error of 95% confidence.

Amisk Group and Metadiorites and Metagabbros

The natural remanent magnetization (nrm) directions from sites at Flin Flon are usually scattered about the Earth's field direction (*e.g.* site 9, Fig. 5(1)). Those from the region immediately east of Amisk Lake are usually well-grouped, generally in southerly directions with variable inclinations (*e.g.* sites 16, 17 and 23, Fig. 5(1)). Some sites show a streaked distribution from southeast to northwest (*e.g.* sites 9, 17 and 23 in part, Fig. 5(1)). Examples of af decay curves are shown in Fig. 6(2). The decay of intensities and the improvement in the grouping of directions indicate the elimination of components imposed by the Earth's present field. The resultant directions after af demagnetization were mainly southerly directed with positive inclination. This is referred to as the **A** magnetization (Table 5, Fig. 7(1)). After demagnetization in higher fields the magnetizations either became too weak or randomly directed. Thermal demagnetization of af cleaned specimens, however, reveals another less-dominant magnetization (called the **B** magnetization) directed either to the northwest or southeast with low inclinations,—thus showing reversals (Table 6, Fig. 7(2)). It is seen most clearly at site 16 (Fig. 5(1) and (2)).

Thermal demagnetization of the **A** magnetization shows that at most sites (18 of 19) the blocking temperatures are mainly between 100 and 400°C but occasionally range up to 500°C. A study of the polished sections revealed the presence of magnetite often with ilmenite lamellae. Ilmenite is the chief opaque mineral in sites 10, 19 and 46. Hematite was identified at several sites showing the **A** direction, usually as an alteration of magnetite. Hematite was not identified in the metadiorites of Flin Flon, though some have coercivities as high as 9000 Oe indicative of hematite. Most sites contain pyrite and pyrrhotite. Many contain extensively

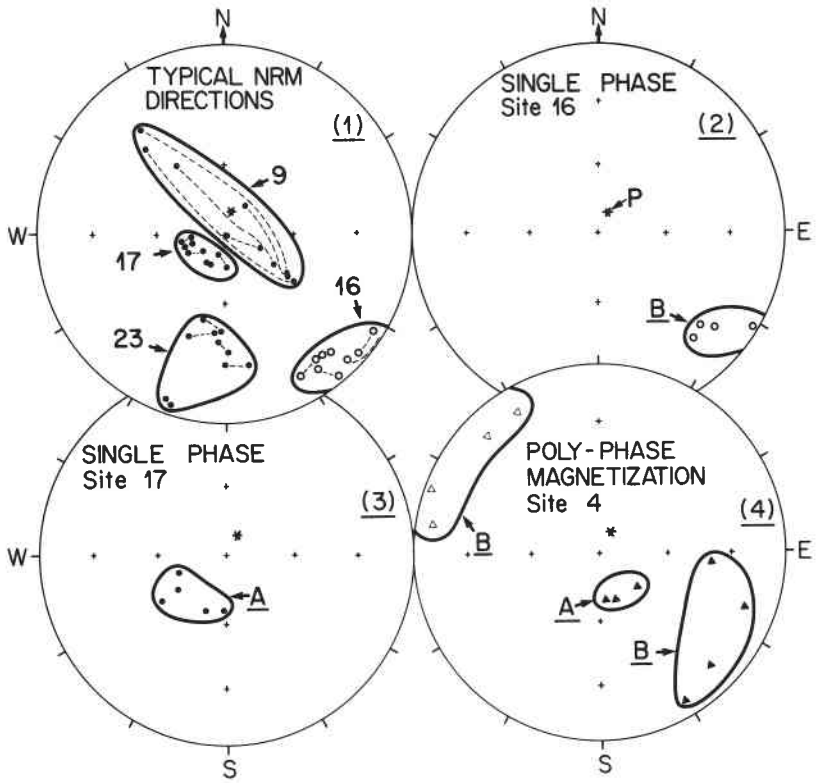


FIG. 5. Examples of directions of nrm and single and polyphase magnetizations. *P* is the direction of the present earth's field. All directions are plotted with respect to horizontal. Solid symbols (open symbols) represent directions plotted on the lower (upper) hemisphere; circular symbols, measured values; and triangular symbols, vector subtraction values (see example Table 3). Protrusions on symbols indicate directions which are reversed 180° from their actual positions.

altered iron minerals with strong internal reflections.

Thermal demagnetization studies of the **B** magnetization (sites 9, 16, and 45) show this direction to be stable from 500 to 600°C. The thermal curves from sites 16 and 45 are square-shouldered and drop off abruptly at 500°C (*e.g.* site 16; Fig. 8(1)). Polished sections reveal ilmenite to be the chief opaque mineral. At site 16 the ilmenite contains patches (up to 15μm), inclusions ($\ll 1\mu$m), and lamellae (1 to 2μm across) of magnetite. At site 45 ilmenite and magnetite form a patchwork mosaic. At site 9 ilmenite occurs as separate elongate grains and as a few intergrowths in magnetite. Hematite was detected at site 45. Magnetite thus appears to be the main carrier of the **B** direction magnetization. The iron minerals are often extensively altered. Site 16 shows ilmenite rimmed with goethite, and this could be the effect of retrograde metamorphism. Sulfides were identified in most sections. The directions of the **A** and **B** magnetizations are summarized in Table 7.

Missi Group

The nrm's show moderate grouping with precisions (k) ranging from 10 to 120. With thermal demagnetization the directions (except for site 27) move away from the Earth's field. Pilot specimens display an optimum precision at 100°C, which thereafter decreases somewhat and then remains approximately constant from 200 to 650°C. Some **A** directions persist up to 675°C. Final site directions are given in Table 5 and sample directions in Fig. 9. The directions are close to the **A** direction of the Amisk Group (Fig. 7(1); Table 7, entry 8). Thermal decay curves are variable with a wide range of blocking temperatures (Fig. 10) but with very stable end-point directions between 600 and 650°C. Polished sections

TABLE 5. Directions of A magnetization

Site	UTMG	M_N	Treatment	N	$D°, I°$ (R)
Amisk Group (Flin Flon)					
3	147738	45.5	200 Oe	3(2)	167, +72(2.76)
7	141712	1.9	500 Oe	4(1)	204, +70(3.96)
8	150710	21.1	150 Oe	3(2)	287, +74(2.98)
9	116693	231.0	100 Oe	4(1)	112, +71(3.87)*
10	118691	4.4	150 Oe	4(1)	185, +78(3.77)
11	124698	4.3	100 Oe	3(2)	264, +75(2.80)
24	188722	62.9	150 Oe	5	149, +71(4.92)
25	166701	1.4	200 Oe	2(3)	166, +61(2.00)
28	154730	6.2	150 Oe	2(3)	163, +44(1.91)
29	157739	0.8	200 Oe	4(1)	237, +80(3.96)
Amisk Group (Amisk Lake)					
17	888606	10.9	250 Oe	5	222, +65(4.90)
19	852492	84.0	150 Oe	4(1)	212, +57(4.00)
20	884603	4.4	200 Oe	4(1)	150, +64(3.96)
21	915602	3910.0	150 200 Oe	5	183, +25(4.95)
23	911641	138.1	250 Oe	5	181, +19(4.58)
46	914636	8.5	200 Oe	2(2)	203, −07(1.95)†
47	917627	0.3	—	(5)	—
Metadiorites and metagabbros (Flin Flon)					
1	148729	1.1	150 Oe	3(2)	148, +67(2.94)
2	151734	6.7	200 Oe	3(2)	181, +65(2.83)
12	127697	0.3	—	(5)	—
13	129710	2.2	200 Oe	5	112, +73(4.90)
15	103695	7.0	300 Oe	3(2)	177, +87(2.91)
36	150695	0.9	250 Oe	4(1)	094, +82(3.76)
44	142736	119.0	150 Oe	5	139, +71(4.59)
Metagabbro (Amisk Lake)					
22	917616	5.3	400 Oe	3	316, +70(2.89)
Amisk-like rocks (Hanson Lake)					
32	340655	0.3	250 Oe	4(1)	152, +76(3.73)
33	347647	170.0	100 Oe	6	135, +68(5.95)
34	353638	85.0	100 Oe	3(2)	150, +68(2.88)
Missi Group (Flin Flon)					
26	178729	14.4	650°C	5	164, +86(4.74)
27	166748	86.6	150 Oe	4(1)	351, +76(3.50)†
38	284661	19.1	650°C	5	199, +69(4.98)
39	284664	12.4	600°C	5	205, +60(4.97)
40	283673	17.9	650°C	5	206, +74(4.97)
41	165708	11.5	650°C	5	179, +78(4.87)
42	284677	11.9	650°C	4(1)	220, +76(3.98)
Boundary Intrusions (Flin Flon)					
4	138738	301.0	200 Oe	2	130, +75(1.99)*
4		—	575°C	4	154, +62(3.87)
5	134746	244.0	500°C	5	343, −61(4.94)*
5		—	600°C	3	122, +48(2.92)
6	139738	43.3	200 Oe	5	185, +85(4.96)
14	176712	2.5	150 Oe	3(2)	152, +70(2.97)*
30	145688	7280.0	150 Oe	6	118, +66(5.70)*
30		—	250–500 Oe	5	256, −78(4.62)*
30		—	550°C, 575°C	3	192, +78(2.96)*†
30		—	575°C, 600°C	3	022, −70(2.93)†
31	144689	73.3	1000 Oe	5	277, −51(4.97)
35	151693	70.6	1000 Oe	5	207, +65(4.99)†
35		—	550°C	5	210, +65(5.00)
37	150696	28.6	750 Oe	5	160, +76(4.98)
43	142737	9.8	150 Oe	5	184, +81(4.94)

NOTES: UTMG is the universal transverse mercator grid reference. All sites except those from Amisk Lake are referred to zone 14 of the Schist Lake topographic map (63K/12E edition 2). Amisk Lake sites are referred to zone 13 of the Denare Beach topographic map (63L/9). M_N is the nrm intensity. N is the number of core samples, with those discarded given in brackets (see text). R is the vector resultant. Other symbols are given in Table 2.

*Direction obtained by vector subtraction (example, Table 2).

†Direction not used in averages.

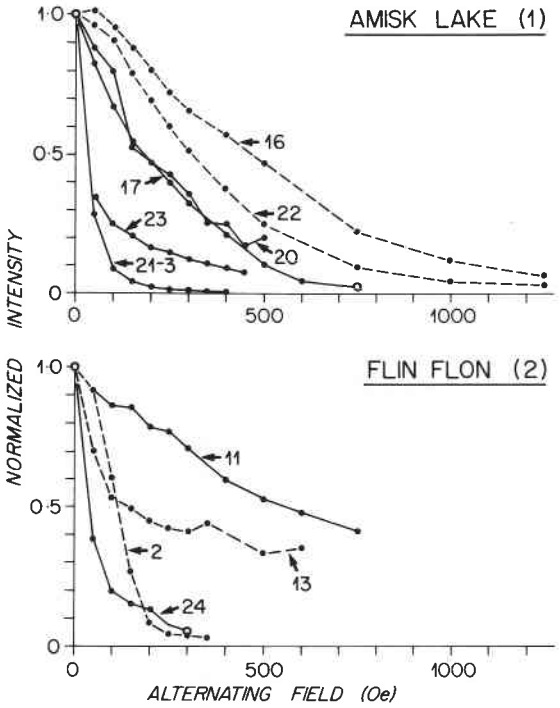


FIG. 6. Typical af decay curves of Amisk specimens (solid curves) and metadiorite and metagabbro specimens (dashed curves) from (1) Amisk Lake and (2) Flin Flon. Solid dots (open circles) denote 'stable' ('unstable') directions. 'Stable' directions remain within about 20° of one another. This convention is used for all succeeding thermal and af curves.

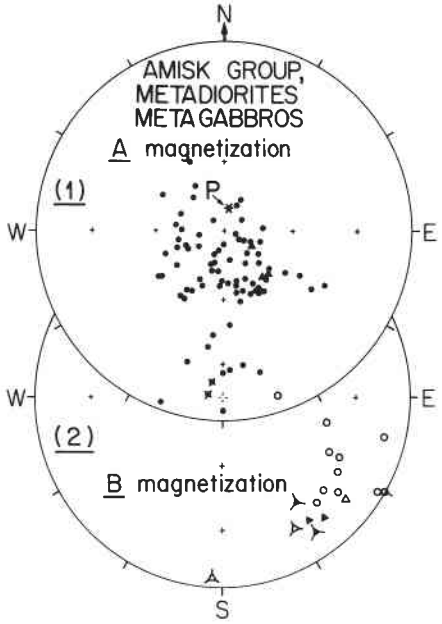


FIG. 7. Cleaned sample directions from the Amisk Group and the metadiorites and metagabbros. Symbols explained in Fig. 5.

show magnetite largely altered to intergrown hematite (typically $<50\mu$m). No **B** magnetization was found.

Boundary Intrusions

The nrm's are usually scattered, but a few sites show good groups away from the Earth's field direction. Af demagnetization produces high-precision **A** directions (Table 5, Fig. 11(1)). There are two 'reversed' sites (5, 31). Af demagnetization curves are shown in Fig. 12(2). With thermal demagnetization, specimens often remain in their initial **A** directions from 500°C to above 550°C, but some move to the northwest (**B** directions) between 400° and 550°C. Several sites with **A** directions show very complex behavior. The directions during demagnetization move away from, and then revert to, the **A** direction. This sometimes occurs after displaying momentarily an intermediate **B** direction (site 4) and sometimes by just reversing (at site 30 this reversing effect occurs three times). The *later-derived* **A** directions are commonly stable up to 625°C. One specimen from site 5 yields an **A** direction which is stable up to 650°C. Thermal decay curves are given in Fig. 12(1). The curves for af treated specimens are more square-shouldered, since the unstable magnetization has been first removed. Blocking temperatures, mainly between 500 and 600°C, indicate the presence of magnetite.

Polished sections from all sites (site 14 excepted) show magnetites ranging up to 1000 μm, and comprising up to 99% (site 30) of the opaque minerals present. Sulfides, present in 5 of the 9 sites, can constitute up to 30% (site 4) of the total opaque minerals. At site 14, the quartz diorite is altered, and no opaque minerals were identified. Ilmenite can comprise up to 90% of the opaque minerals (site 35), and is associated with magnetite as intergrowths, or adjacent grains. Hematite was identified at sites 30, 31 and 43. At sites 30 and 31 it appears to be an alteration product of magnetite. All sites contain many small opaque minerals usually impregnating the silicates, but these could not be identified. At site 30, opaque minerals, believed to be magnetite, are found surrounding or between silicate grains, and in the center of material filling large fractures. The infilling

TABLE 6. B Direction magnetization

Site	Fm	UTMG	M_N	Treatment	N	$D°, I°(R)$
9	Am	116693	—	500°C	4	335, −05(3.60)*
9	Am		—	600°C	3	137, +09(2.86)*
16	Mg	901625	4.6	500 Oe	5	128, −15(4.87)
18	Mg	886580	0.4	—	1(4)	131, +13†
45	Mg	900624	56.2	750 Oe, 550°C	5	113, −29(4.89)
4	Bi	138738	—	200 Oe	4	124, +22(3.69)*
4	Bi		—	550°C	4	304, −09(3.74)*
14	Bi	176712	—	500°C, 550°C	5	300, −25(4.88)

NOTES: The formations (Fm) are respectively: Am Amisk Group (Flin Flon); Mg metagabbros (Amisk Lake); and Bi Boundary Intrusions. Other symbols and superscripts are explained in Table 5. The two results of sites 4 and 9 are treated separately.

material cuts small pre-existing fractures in the silicates. Hence, there appear to be at least two generations of magnetite represented. An olivine pseudomorph altered to iddingsite, containing magnetite stringers, was tentatively identified in site 30.

The opaque mineralogy at sites 35, 37, and 43 is of particular interest. Magnetite grains are partially or wholly altered. The unaltered central portions of grains are darker in color. The altered outer portion, which is often faintly anisotropic—perhaps indicative of fine hematite—contains numerous small dark patches ($\ll 1\mu$m). The central darker portion is possibly titanomaghemite and the outer portion possibly titanohematite. Such minerals, which have been described in Larson *et al.* (1969), are evidently the product of low temperature oxidation. At other Boundary Intrusion sites there is a general lack of high oxidation products and very few lamellae of ilmenite. This interpretation, that the Boundary Intrusions have undergone extensive low temperature alteration, is supported by the thermal demagnetization curves which range from square-shouldered curves (site 35) characteristic of magnetite, to curves (site 30) with widely distributed blocking temperatures.

Amisk-like Rocks from Hanson Lake

Two sites (33, 34) had strong nrm's and the other (32) was quite weak. After af cleaning the sites show an **A** direction (Tables 5 and 7). Examples of af decay curves are shown in Fig. 13(2) and sample directions in Fig. 14. Thermal demagnetization curves from sites 33 and 34 show rapid decay (Fig. 13(1)), but that from site 32 is square-shouldered and shows blocking temperatures between 600 and 650°C. The former sites retain their af cleaned direction from 300 to 500°C, but become unstable past 600°C. The high initial remanences, the comparatively low coercivities, and the scatter of directions at 600°C, would indicate that magnetite is the carrier of the stable remanence at sites 33 and 34. The low initial remanence, higher coercivities, and blocking temperatures above 600°C, indicate that hematite is probably responsible for the stable remanence at site 32. Sites 33 and 34 with 1% opaque minerals contain large irregular to euhedral grains of magnetite (up to 1000 μm) with fine intergrowth textures. These grains often have slight hematite alteration and are rimmed with goethite. They also contain partially altered ilmenite. Site 32 is highly altered with about 95% of its opaque minerals in the form of dull greenish grains. The other opaque minerals are sulfides and small grains—possibly hematite.

Puzzling High Temperature Effects

During this study a few samples exhibited directions with very high blocking temperatures ranging from 675 to 735°C (higher temperatures could not be obtained). These magnetizations (denoted **C**) were usually northerly with shallow to medium inclinations (e.g. Fig. 15). J_s–T curves from three of these samples (Fig. 16) confirm the presence of very high Curie temperatures. The heating curves of the Amisk specimens (Fig. 16) show a distinctive drop between 500 and 550°C near the Curie point of titanomagnetite, and that of the Missi specimen (Fig. 16) a barely perceptible dip at 580°C, the Curie point of magnetite. All three curves show a change of slope near the Curie point of hematite. All curves are irreversible on cooling. In igneous rocks large-scale irreversible changes like these are often associated with samples

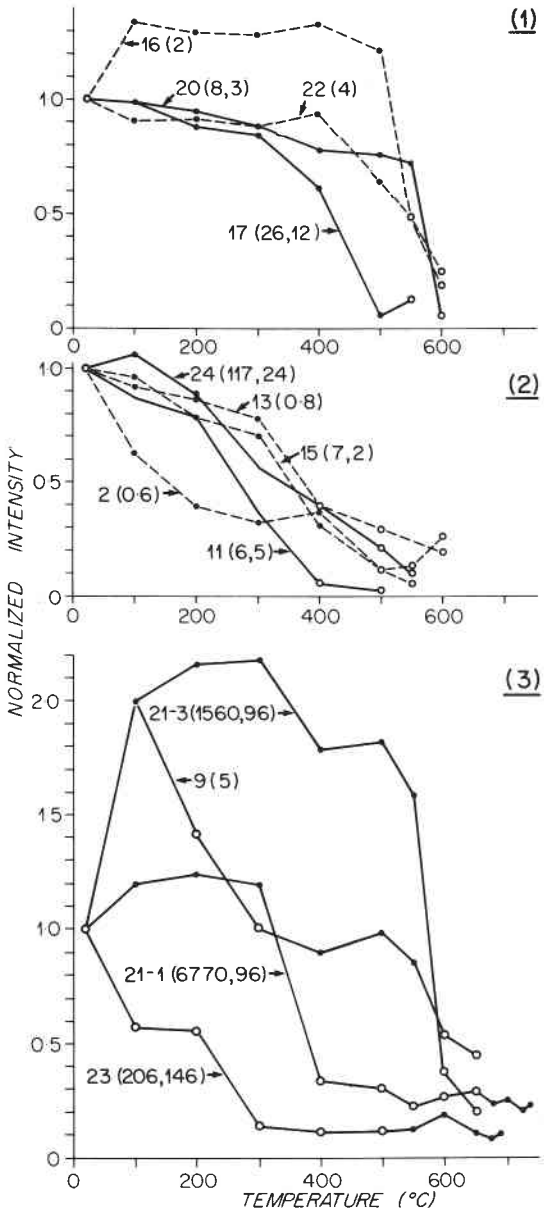


FIG. 8. Typical thermal decay curves of Amisk specimens (solid curves) and metadiorite and metagabbro specimens (dashed curves) from (1) Amisk Lake, (2) Flin Flon, and (3) both Amisk Lake and Flin Flon. The numbers in brackets are: first, the initial intensity, and second, the af cleaned intensity; both given in microgauss.

containing titanium-rich titanomagnetite or low-grade titaniferous maghemite, minerals associated with low temperature oxidation processes (Larson *et al.* 1969). Indeed, the

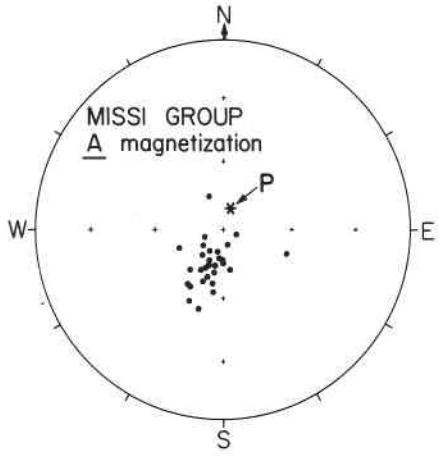


FIG. 9. Cleaned sample directions from the Missi Group. Symbols explained in Fig. 5.

heating and cooling curves of Fig. 16(2) show a distinct, but broad dip, between 100 and 400°C, apparently indicative of extensive low-temperature oxidation (Larson *et al.* 1969). The more blocky aspect of the curve from site 21 (Fig. 16(1)) shows that the low temperature oxidation to titanomaghemite is more complete. The cause of these high blocking temperatures, which range from 675 to 735°C can only be surmised. Native iron (Curie temperature 780°C) is a possibility. Iron has recently been detected in Newfoundland ophiolites by Deutsch and Rao (1973). No iron has been detected in the present polished sections, but it could exist below the resolving power of the microscope. If iron is present, it must be the result of extreme reducing conditions. The presence of abundant sulfides in these rocks indicates that a reducing environment did exist.

Iron contamination caused by drilling or grinding is another possibility. Drilling, using similar bits, has not caused observable contamination of other paleomagnetic collections. Powders ground for the above Curie point determinations have confirmed the high blocking temperatures indicated by thermal demagnetization. These powders were prepared with the specific intention of avoiding iron contamination. Therefore iron contamination seems to be an unlikely explanation.

There are however reasons for believing that the high temperature directions do not record the paleomagnetic field. Readings, made on a Schonstedt spinner magnetometer, had a ten-

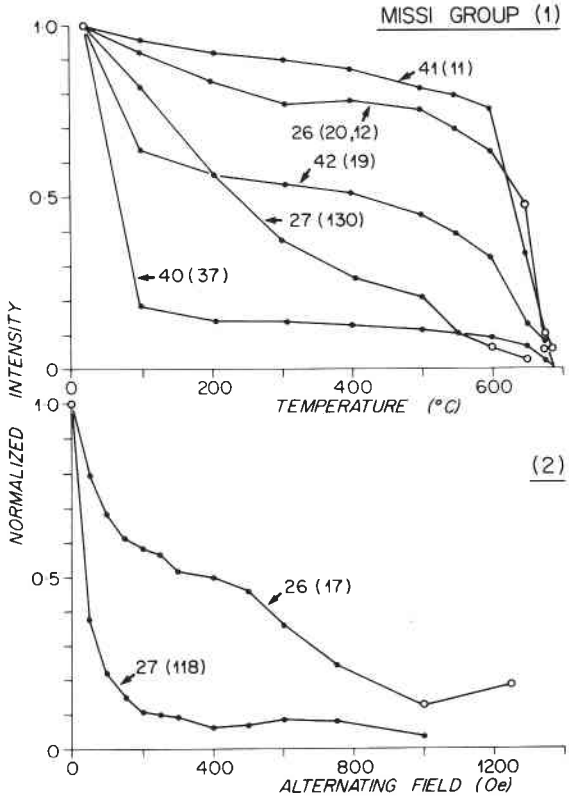


FIG. 10. Typical thermal (1) and af (2) decay curves of Missi Group specimens. See legend of Fig. 8.

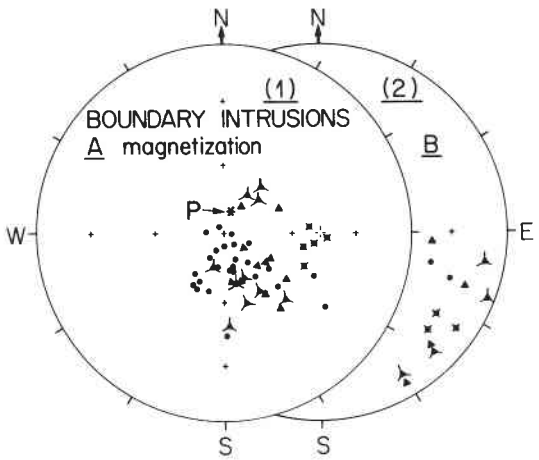


FIG. 11. Cleaned sample directions from the Boundary Intrusions. Symbols explained in Fig. 5.

dency to change gradually whilst the specimen was spinning, and to be consistently higher in two of the measurement positions. Moreover measurements made with an astatic magnetometer (Roy *et al.* 1972) showed the existence of large induced components when specimens were positioned at distances of 5.1 to 6.1 cm below the lower magnet of the suspension system. These measurements resulted in steep directions of magnetization in contrast to the shallower directions produced by the Schonstedt. Thus the measurement problems associated with these specimens are not resolved, and no reliance can at present be placed on them.

Discussion and Summary

All the rock units contain **A** magnetization. **B** magnetization occurs in all units except the Missi Group.

The **A** magnetization is tentatively divided into three categories. **A1** has low blocking

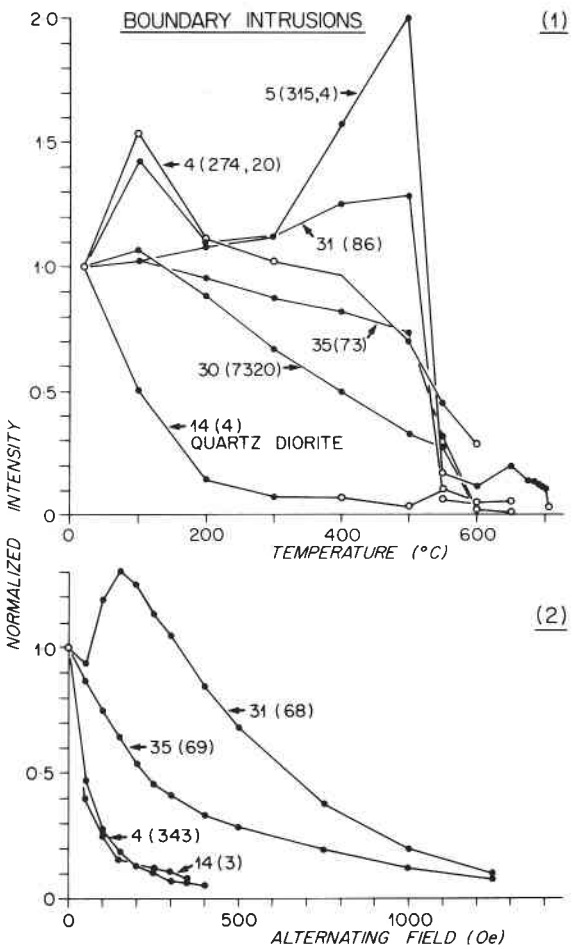


FIG. 12. Typical thermal (1) and af (2) decay curves of Boundary Intrusion specimens. See legend of Fig. 8.

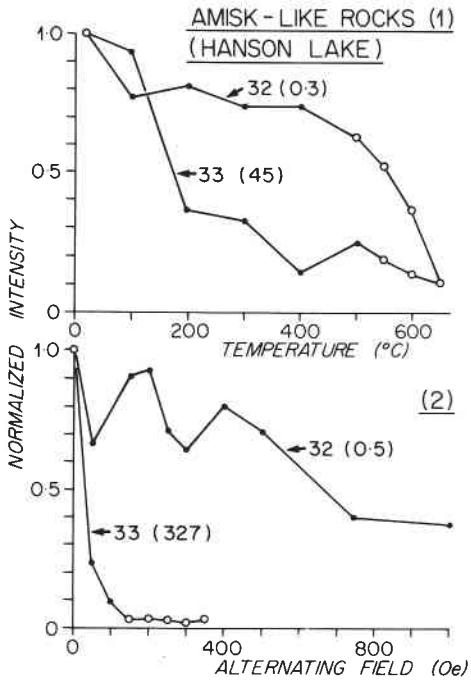


FIG. 13. Typical thermal (1) and af (2) decay curves of Amisk-like rocks from Hanson Lake. See legend of Fig. 8.

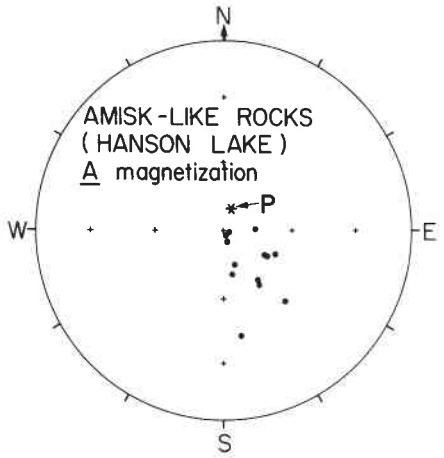


FIG. 14. Cleaned sample directions from the Amisk-like rocks. Symbols explained in Fig. 5.

temperatures (<500°C; Fig. 17(1), (2)), coercive forces in the lower mcf range (Fig. 4(2)), and is found in all rock units. It is very probably due to magnetite and is by far the dominant magnetization at Flin Flon. **A2** has blocking temperatures between 550 and 625°C (Fig. 17(4)), coercive forces generally in the higher mcf

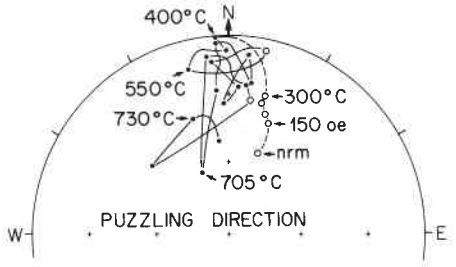


FIG. 15. Example of a typical high-temperature direction. Symbols explained in Fig. 5.

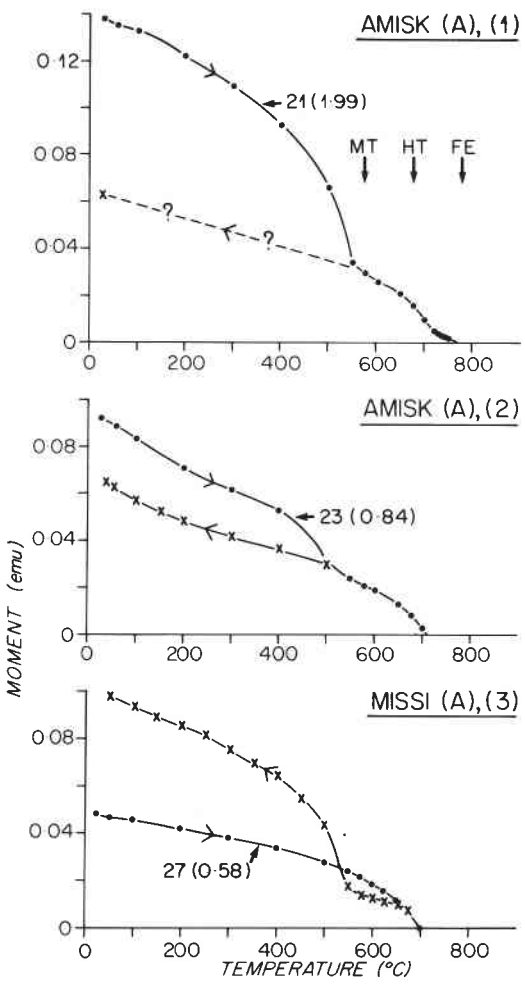


FIG. 16. Examples of J_s–T curves of Amisk (1 and 2) and Missi (3) specimens. The heating (cooling) curve is denoted by solid dots (crosses). The value in brackets is the initial intensity in emu/g. MT, HT, and FE indicate the Curie points of magnetite, hematite, and iron respectively.

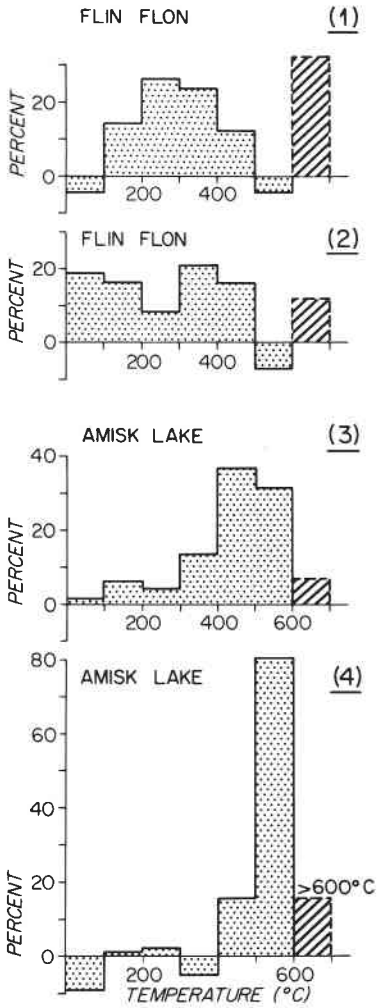


FIG. 17. Blocking temperature spectra of Amisk and metadiorite-metagabbro specimens combined. (1) and (3) are for 6 and 3 specimens respectively cleaned in af before thermal demagnetization. (2) and (4) are for 10 and 3 specimens respectively without previous af demagnetization. All spectra were obtained from normalized curves. That portion of the spectra >600 °C represents largely a noise component.

range (Fig. 4(2)), and is probably due in part to magnetite and in part to hematite. **A2** is usually evident after further cleaning of the **A1** magnetization, and shows reversals. **A2** has been found in the Boundary Intrusions. It may occur in the Amisk rocks at Amisk Lake, but not enough data were obtained for its resolution through vector analysis. **A3** has blocking temperatures up to 683°C, coercive forces in the hcf range (Fig. 4(2)), and is probably due to hematite. It has been found only in the Missi Group.

The three **A** magnetizations have some statistical significance (Table 7). The error circles (α_{95}) of **A2** and **A3** do not overlap, and the means therefore differ significantly. The error circle of the **A1** direction overlaps the error circles of the **A2** and **A3** magnetization. Further statistical analysis shows however that **A1** and **A2** differ significantly at the $P = 0.5$ level. Thus there are statistical grounds for believing that the three **A** magnetizations are distinct.

The **B** magnetization is usually, but not always, more stable versus af fields than the **A1** magnetization, and this is why the two cannot be clearly separated by af demagnetization alone. At sites 16 and 45 **B** is very stable indeed. The **B** magnetization has blocking temperatures above 500°C (Fig. 17(4)), and probably corresponds to the high mcf values of Fig. 4. It is probably due to more or less pure magnetite. The presence of reversals shows that **B** is not grossly biased by an unremoved magnetization. It is therefore suggested that it was acquired prior to the **A** magnetizations.

It is suggested that the **A1** magnetization is a viscous partial thermoremanent magnetization (vptrm) developed at low temperatures over a long period and stabilized by cooling during uplift, whereas the **A2** and **A3** magnetizations are chemical remanent magnetizations (crm's) acquired mainly through the secondary development of hematite. **A3** is thought to be produced during retrograde metamorphism associated with faulting during P3 deformation or with post-up lift fault movement. The suggestion that the **A** magnetizations were acquired at low-temperature is supported by the opaque mineralogy and J_s–T curves, which indicate low-temperature oxidation. It is suggested that the **B** magnetization is an original thermoremanent magnetization (trm) in the Boundary Intrusions, but is a vptrm in other rocks caused by heating during the emplacement of the Boundary Intrusions.

The poles corresponding to the **A** and **B** directions are plotted in Fig. 18 and listed in Table 7. The three **A** magnetizations have poles which apear to be significantly different from one another (see above). The **A1** pole (Table 7, entry 12) lies close to the mean **A** pole (entry 10). **A2** may pre-date **A1**, because its pole

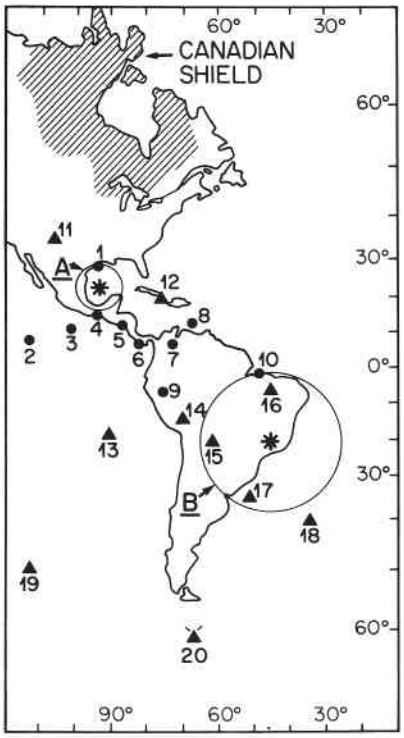


FIG. 18. Pole map. Poles of the **A** and **B** directions are shown with their error α°_{95} circles. Poles from other studies from that part of the Canadian Shield (shaded) stabilized before about 1600 m.y. ago are also shown. Dots denote poles in the approximate time range of 1600 to 2000 m.y. and triangles poles in the time range >2000 m.y. The poles are as follows: (1) Gunflint Formation (Symons 1966), (2) Western Channel diabase (Irving *et al.* 1972*a*), (3) metamorphic rocks, Daly Bay (Park 1973), (4) Kahochella Group *S* (McMurray *et al.* 1973), (5) Nonatcho Group sediments (McGlynn *et al.* 1974), (6) Dubawnt Group (Park *et al.* 1973), (7) Kahochella Group *P* (McMurray *et al.* 1973), (8) Sparrow dikes (McGlynn *et al.* 1974), (9) Martin Formation (Evans and Bingham 1973), (10) Et-Then Group (Irving *et al.* 1972*b*), (11) Otish Gabbro (Fahrig and Chown 1973), (12) Indin dikes (McGlynn and Irving, in press), (13) Nipissing diabase (Patel and Palmer 1974), (14) Nipissing diabase (Symons 1970), (15) Ex dikes (McGlynn and Irving, in press), (16) Duck Lake sill (McGlynn and Irving, in press), (17) Dogrib Dikes (McGlynn and Irving, in press), (8) dike from Beartooth Mountains (Larson *et al.* 1973), (19) Matachewan diabase (Strangway 1964; Fahrig *et al.* 1965), (20) Stillwater Complex, Montana (Bergh 1970).

(Table 7, entry 14), though calculated from only four sites, lies 36° to the east of **A1**. The pole of **A3** lies 16° west of **A1**. The probable age sequence is from oldest to youngest **A2**, **A1**, **A3**, but this is only a tentative suggestion.

The age of the **A** pole is interpreted to be between 1600 and 1700 m.y., since the K–Ar method probably dates the last stages of cooling associated with uplift, and the magnetization is considered to be related to this cooling. The **A** pole is located among poles mainly dated at about 1600 m.y., and this is consistent with the above age assignment. The **A2** pole falls close to the pole for the Sparrow dikes, which have a probable age of about 1700 m.y. (Fig. 18, pole 8), and the **A3** pole is not far from the pole for the Daly Bay metamorphic rocks dated at 1620 m.y. (pole 3, Fig. 13).

All the evidence available at present is consistent with the idea that the **B** magnetization is older than **A**. It presumably is to be related to the Rb–Sr ages for which a mean of 1800 m.y. may be cited (Table 1). The pole associated with the puzzling high-temperature **C** magnetization is at 45°S, 088°W, and lies about 20° east of the Matachewan pole (Fig. 18, pole 19) which is dated at 2690 m.y. All other early Proterozoic and Archean poles from the Canadian Shield are plotted in Fig. 18 for comparison. It is obvious that the interpretation of these poles is not yet clear, and the above suggestions are tentative only.

If, as has been argued above, the **B** magnetization of the Boundary Intrusions is a trm, then it is either late- or post-P3 in age. Since the **B** pole is very different from the **A1** and **A2** poles which seem to be associated with uplift following P3, it is reasonable to suppose that **B** is late-P3 in

ROCK UNIT	TECTONIC EFFECTS	POSSIBLE AGE (m.y.) *	MAGNETIZATION
	late fault movements ?	<1600	A 3
	uplift	1600 - 1800	A1 + A2
Boundary Intrusions granitic rocks	P3	1800 - 2000	B
granitic rocks		>2000	
granitic rocks	P2		C??
	P1		
Missi Group			
Amisk Group	weak deformation		

* inferred from pole positions (Fig. 18)

FIG. 19. Tentative correlation of the magnetic record with the tectonic history. P1, P2, and P3 are the three phases of deformation of Stauffer and Mukherjee (1971). *C*? is the very tentative high-temperature magnetization.

TABLE 7. Summary of directions and poles

Entry	Description	N	$D°, I°$	k	$\alpha°_{95}$	Pole
1.	**A** Amisk Group (Flin Flon)	10	179, +75	20	11	27N, 101W
2.	**A** Amisk Group (Amisk Lake)	5	188, +48	9	27	06S, 109W
3.	**A** Amisk Average (1 and 2)	15	184, +67	11	12	14N, 105W
4.	**A** Metadiorites (Flin Flon)	6	145, +76	47	10	31N, 084W
5.	**A** Metagabbro (Amisk Lake)	1	316, +70	—	—	64N, 176W
6.	**A** Extrusives, Intrusives (1, 2, 4, 5)	22	178, +72	12	9	21N, 101W
7.	**A** Amisk-like rocks (Hanson Lake)	3	145, +71	212	9	24N, 081W
8.	**A** Missi Group (=**A3**)	6	201, +74	71	8	27N, 113W
9.	**A** Boundary Intrusions	12	140, +72	23	9	26N, 079W
10.	**A** Grand Average (1, 2, 4, 5, 7, 8, 9)	43	168, +73	17	5	24N, 095W*
11.	**A** Grand Average, Poles	43	—	8	8	27N, 095W
12.	**A1** Directions	33	171, +73	17	6	23N, 097W
13.	**A1** Poles	33	—	8	9	26N, 096W
14.	**A2** Directions	4	117, +62	17	23	22N, 057W
15.	**A2** Poles	4	—	9	33	24N, 059W
16.	**A3** Poles (see 8)	6	—	27	13	28N, 113W
17.	**B** Directions	7	129, +04	12	18	20S, 046W*
18.	**B** Poles	7	—	22	13	21S, 050W

NOTES: The magnetization category is indicated in bold-face. **A1** includes the **A** site directions of the Amisk Group, the metadiorites and metagabbros, the Amisk-like rocks from Hanson Lake, and all the Boundary Intrusions except those entries which are **A2** directions (Table 5: site 31 and the second entries of sites 4, 5, and 30). Symbols are explained in Tables 3, 5 and 6.
*The semi-axes of the polar errors (δm, δp) of entries 10 and 17 are (10°, 9°) and (18°, 9°) respectively.

age. The **A3** pole, from its position and from its field association with young fault zones, is interpreted to be due to renewed fault movement. Unless one admits the reality of the puzzling high-temperature directions, which this author is not prepared to do, there is no evidence that any of the magnetizations are P2 or older in age. This tentative interpretation is set out in Fig. 19. There is no firm evidence for pre-Aphebian magnetization. This does not mean that Archean rocks are absent, but only that if present they have been remagnetized. This work does not therefore provide an answer to the problem of the age of the Amisk Group.

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Appendix

In this study a specimen from each collecting site was given a saturation irm and demagnet-

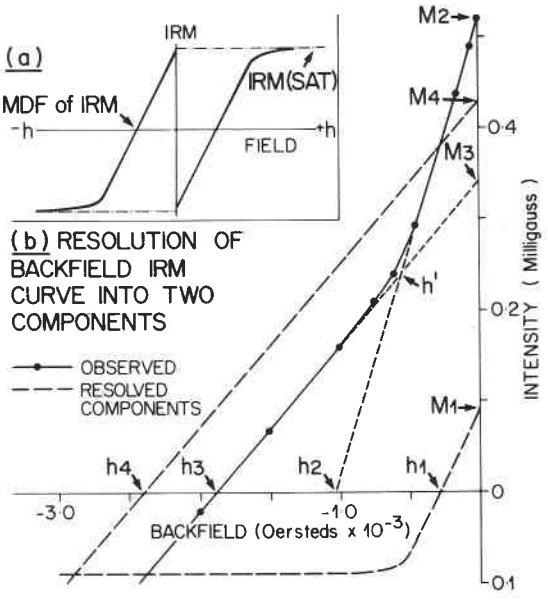


FIG. 20. Resolution of curves of the backfield demagnetization of irm. (a) Ideal single-component curve showing complete hysteresis loop. (b) Two-component curve resolved. The dots are experimental points.

ized by a backfield. The backfield demagnetization curves of most Flin Flon rocks show a significant change of slope between roughly straight line segments (Fig. 20(b)). The changes of slope appear to indicate the presence of magnetizations with different remanent coercive force spectra. The problem therefore is to determine the coercivities of these different magnetizations present. As a first approximation it is assumed that each component is completely saturated and has a linear decay. Such an ideal one-component remanence hysteresis curve is illustrated in Fig. 20(a). An actual example of a curve resolved into two components with coercivities h_1 and h_4 is given in Fig. 20(b). Values for h_1 and h_4 can be calculated from these formulae

$$h_1 = \frac{h_2 h_3 (M_2 - M_3)}{2(h_3 M_2 - h_2 M_3)}$$

$$h_4 = \frac{h_3 (M_2 + M_3)}{2M_3}$$

M_2 is the observed saturation irm, and h_3 the apparent (observed) mdf or remanent coercivity. M_4, h_4, and M_1, h_1 are the saturation irm and coercivity of the magnetizations with high and low remanent coercivity. M_3, h_2 are obtained by construction (Fig. 20).